Experiment No. 25 (Mixture No. 3)

Date: / /

Aim : Analyse two basic (cation) radicals qualitatively from given inorganic mixture.

Apparatus : Test tubes, test tube holder, test tube stand, filter paper etc.

A. Preliminary tests:

Test	Observation	Inference
Color	white	Br^{2+}, Ba^{2+} may be Present
Nature	Crystalline	water Soluble Salts Present.

B. Dry Tests for Basic radicals (Test tube must be dry for this test)

Test	Observation	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture a clean and in dry test tube and heat it strongly.	i) cracking noise ii) formation of white sublimate	crystalline salts like $Pb(NO_3)$ NaCl, KBr etc may be present. NH_4^+ may be present
2. Charcoal Cavity Test: Mixture + Na_2CO_3 solid in 1:2 proportion placed in fresh charcoal cavity, moisten with a drop of water. Heat it with blow pipe in a reducing (yellow) flame	Substances fuses & sinks in the cavity	NH_4^+ or Ca^{2+} may be present
3. NaOH Test: Mixture + NaOH solution & heat. Hold moist turmeric paper near the mouth of the test tube	Moist turmeric paper turns brown/red	NH_4^+ may be present.
4. Flame Test: Prepare a paste of the given mixture with conc. HCl on a watch glass. Make a small loop at the end of the platinum wire & dip it in the mixture or use glass rod. Heat it on oxidising flame (Blue) observe the colour change of the flame.	Green flashes	Zn^{2+} and Mn^{2+} salts may be present

C. Preparation of original solution (O.S)

Take a small quantity of mixture in a beaker add 20 mL of distilled water, stir with glass rod to dissolve the mixture. If mixture does not dissolve completely then warm it to dissolve. Clear solution is obtained, which is used as a O.S for further tests.

1. Analysis of Group zero (NH_4^+)

Test	Observation	Inference
1. O.S. + NaOH solution + Heat, test with moist turmeric paper.	i) Evolution of NH_3 gas which turns mois turmeric paper brown/red	Group zero present (NH_4^+)

2. O.S. + NaOH solution + Heat. Bring a glass rod dipped in conc. HCl. near the mouth of the test tube.	Dense white fumes of NH_4Cl	NH_4^+ present
C.T. for NH_4^+ : i. O.S. + Nessler's reagent in excess	Brown ppt / colousation	NH_4^+ confirmed
ii. O.S + Picric Acid (2, 4, 6 trinitro phenol)	yellow ppt	NH_4^+ confirmed

Detection of group (if group zero is absent, two groups must be detected following test)

Test	Observation	Inference
1. O.S. + dil. HCl	No white ppt	Group I absent
2. O.S./Filtrate + dil HCl (heat) + H_2S gas or water	No Black ppt	Group II absent
3. O.S./Filtrate (Remove H_2S) + NH_4Cl (equal) + NH_4OH (till alkaline to litmus)	no ppt	Group III absent
4. O.S./Filtrate + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + H_2S gas or water	white / Black / fresh of Boff ppt	Group IV (Mn^{2+}, Zn^{2+} Co^{2+}, Ni^{2+}) present
5. O.S./Filtrate (Remove H_2S) + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + $(NH_4)_2CO_3$	—	—
6. O.S./Filtrate + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + Na_2HPO_4	—	—

2. Analysis of first detected group Group -zero- (NH_4^+)

Test	Observations	Inference
i) O.S + NaOH sol^n + heat & cold moist turmric paper near the mouth of test tube.	Evolution of NH_3 gas which turns moist turmeric paper brown/red	Group zero present (NH_4^+)
ii) O.S + NaOH sol^n + heat Bring a glass rod dipped in conc. HCl near the ~~mouth the test tube~~	Dense white fumes of NH_4Cl	NH_4^+ present
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3. C.T. for first detected radical NH_4^+

	Test	Observations	Inference
1.	O.S + Nessler's reagent in excess	Brown ppt/colouration of basic mercury(II) amido Iodine	NH_4^+ confirmed
2.	O.S + Picric acid (2,4,6 trinitrophenol)	yellow ppt of ammonium picrate	NH_4^+ confirmed

2. Analysis of second detected group Group (IV) (Mn^{2+}, Zn^{2+}, Co^{2+}, Ni^{2+})

	Test	Observations	Inference
1.	O.S/filtrate/NH_4Cl + NH_4OH (till alkaline) + H_2S gas or water	White ppt	Zn^{2+} present
2.			
3.			

3. C.T. for second detected radical Zn^{2+}

	Test	Observations	Inference
1.	Above Soln + NaOH	white ppt soluble in excess of NaOH but reprecipited by H_2S	Zn^{2+} Confirmed
2.	Above Solution + $K_4[Fe(CN)_6]$	white or bluish ppt	Zn^{2+} confirmed

Result:-

The given inorganic mixture no. 3 contains following two cations (Basic Radicals)

i) NH_4^+ (Name of cation) Amonium - cation

ii) Zn^{2+} (Name of cation) zinc - cation

Remark and sign of teacher: ....